

# Project 1 — Team Proposal (Refreshed, Milestone 1)

**Team:** Abdellah Faleh, Jin-Woo Hong, Samuel Tong, Tristan Lyons

**Course:** EI 7615 — Deep Learning Architectures

**Date:** September 26, 2026

## Project

Discriminative computer-vision pipeline for celebrity identification and detection (Project 1). Milestone 1 establishes a single-face classification baseline over a small CelebA identity subset; Milestones 2-3 extend it to multi-face detection with YOLOv8.

## Celebrity identities (final list)

Selected 4 identities — the CelebA IDs each team member individually claimed for the class-wide identity-pooling task (23-25 images each), rather than an independently-diversity-selected set:

| Identity ID | # Images | Claimed by     | Dominant attributes |
|-------------|----------|----------------|---------------------|
| 3           | 25       | Abdellah Faleh | Male, Wearing_Hat   |
| 7           | 24       | Jin-Woo Hong   | Male, Black_Hair    |
| 1212        | 25       | Samuel Tong    | Male, Black_Hair    |
| 8335        | 25       | Tristan Lyons  | Male, Brown_Hair    |

## Framework choice

**PyTorch** (torchvision for the pretrained ResNet18 backbone). Chosen for: flexible custom nn.Module definitions for the from-scratch CNN; torchvision.models gives ImageNet-pretrained weights for the transfer-learning baseline with feature-extraction and fine-tuning modes built in; runs on both CPU (Northeastern Explorer) and any team member's local GPU without code changes.

## Division of labor

| Task                                                                                                         | Owner          |
|--------------------------------------------------------------------------------------------------------------|----------------|
| Data pipeline, model training (Custom CNN + ResNet18)                                                        | Abdellah Faleh |
| Results write-up (proposal, training-results docs)                                                           | Abdellah Faleh |
| GitHub repo setup & maintenance, reproducibility (README, environment)                                       | Jin-Woo Hong   |
| Evaluation & review (results, confusion matrix/metrics sanity-check, proposal review)                        | Samuel Tong    |
| Milestone 2 planning (synthetic multi-celebrity dataset, YOLO annotations), Canvas submission & coordination | Tristan Lyons  |

## Anticipated risks

- **Compute constraints:** Explorer's CPU-only nodes (some without AVX2) can silently kill TensorFlow/PyTorch kernels during model-building ops; mitigated by keeping image size (128x128) and batch size small, and by testing on a different partition/node if a kernel dies with no traceback.
- **Class imbalance:** CelebA identities have uneven image counts; mitigated by selecting identities within a bounded image-count range and stratifying the train/val/test split per identity.
- **Small per-identity sample size:** only 23-25 images per identity limits how much a from-scratch CNN can learn. This risk materialized during training: an initial custom CNN configuration (heavier augmentation, higher learning rate) collapsed to predicting a single class (20% accuracy, equivalent to random guessing). We mitigated it by lightening augmentation, lowering the learning rate, and reducing dropout, which brought the custom CNN to a genuine (if lower, on this harder

identity subset) 50% test accuracy — below the fine-tuned ResNet18's 87.5%, but well above the 25% random-guessing baseline. See docs/training\_results.md for the full comparison.

- **Milestone 2 dependency:** the model carried forward from Milestone 1 becomes part of the Milestone 2/3 detection pipeline, so architecture/checkpoint format decisions made here should stay compatible with a YOLOv8 integration later.